LifeBridge Smart Organ Transplant System

A Mini Project Report Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE ENGINNERING

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DECLARATION

The Mini Project Report entitled **"LifeBridge Smart Organ Transplant System"** is a record of Bonafide work of **G. Rithwik Sai – 2420030415, A. Nikitha – 2420030505, V. Sraavya – 2420030584** submitted in partial fulfillment for the award of B. Tech in Computer Science and Engineering (or) Computer Science and Information Technology to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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CERTIFICATE

This is certify that the mini project based report entitled **"LifeBridge Smart Organ Transplant System"** is a bonafide work done and submitted **G. Rithwik Sai–2420030415, A .Nikitha–2420030505, V. Sraavya–2420030584** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Department of Computer Science Engineering, K L (Deemed to be University), during the academic year **2026-2027.**

Signature of the Course Co-ordinator **Signature of HOD**

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1. ABSTRACT

LifeBridge Smart Organ Transplant System is a service-oriented healthcare platform designed to manage and coordinate organ donation and transplantation activities. The system aims to simplify the management of donors, recipients, available organs, matching, allocation and notifications through a single platform. It uses a microservices architecture, where each service handles a specific function independently. This approach improves modularity, maintainability and scalability. The proposed system includes secure user access, donor and recipient management, organ availability tracking and compatibility-based matching support. It is planned using Java, Spring Boot, REST APIs and MySQL for backend development and database management. The system will also support communication between services and provide notifications for important updates. LifeBridge aims to reduce manual coordination, organize transplant-related information and improve the efficiency of the overall workflow. Future enhancements may include AI-assisted matching, mobile application support, analytics and integration with hospital systems. The project is intended to provide a structured and secure approach to organ transplant management.

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Team Members:

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2. **A. Nikitha** – System Integration & Testing
3. **V. Sraavya** – Database & Security Developer

2. INTRODUCTION

Organ transplantation is an important medical process that helps save the lives of patients with failing organs. Managing organ donation and transplantation involves coordinating donors, recipients, hospitals and transplant coordinators. Handling this information manually can lead to delays, difficulties in tracking organ availability and communication problems.

LifeBridge Smart Organ Transplant System is a proposed healthcare platform designed to simplify and organize the organ transplantation process. It manages donor and recipient details, tracks available organs, supports compatibility-based matching and allocation, and provides notifications for important updates.

The system follows a **microservices architecture**, where each function is handled by a separate service. This makes the system modular, easier to maintain and scalable. The proposed technologies include Java, Spring Boot, REST APIs and MySQL, with security features to control access to sensitive information.

The main aim of LifeBridge is to bring important transplant management activities into one organized platform. Future enhancements may include AI-assisted matching, mobile application support and integration with hospital systems.

3. LITERATURE SURVEY

1. Online Organ Donation Management System

This project focuses on managing organ donation activities through a web-based application. It maintains donor and recipient information and helps organize organ availability and matching details. The system aims to reduce manual work and improve communication between the people involved in organ transplantation. It provides a digital approach to managing organ donation records.

Reference: https://www.ijirmps.org/research-paper.php?id=230450

2. UNOS Organ Matching System

The United Network for Organ Sharing (UNOS) uses a computerized system to match donated organs with suitable transplant candidates. The matching process considers factors such as blood type, medical urgency, organ size and waiting time. This system helps organize

organ allocation and supports fair and efficient transplantation procedures. It serves as a reference for designing the matching process in LifeBridge.

Reference: https://unos.org/transplant/how-we-match-organs/

3. Smart Blockchain-Based Organ Donation System

This research explores the use of blockchain technology in organ donation management. Blockchain can help maintain transparent and traceable records and improve data integrity. The study highlights the importance of security and reliability in managing organ donation information. These concepts can help guide future security improvements in LifeBridge.

Reference: https://doi.org/10.51584/IJRIAS.2025.100500092

4. Streamlining Organ Transplantation

This project focuses on improving organ transplantation through a web-based system for managing donor and recipient medical information. It explores methods for identifying suitable organ matches based on relevant medical details. The approach aims to reduce manual coordination and make the matching process more organized. It provides useful ideas for developing the matching and allocation modules in LifeBridge.

Reference: https://www.ijsat.org/research-paper.php?id=5577

5. AIDONOR: Offline Organ Donation Management System

AIDONOR explores an organ donation management approach designed to support environments with limited internet connectivity. It focuses on donor registration, compatibility-based matching and notifications. The system aims to make organ donation information more accessible and improve coordination. Its approach provides useful ideas for developing reliable organ management services.

Reference: https://doi.org/10.1109/IC2NC67409.2025.11376432

6. NOTTO Transplant Manual

The National Organ and Tissue Transplant Organization (NOTTO) provides guidelines related to organ transplantation in India. The manual explains important aspects of organ allocation, compatibility, priority and coordination among transplant organizations. These guidelines help establish the requirements for a structured organ transplant management system. LifeBridge can use these guidelines as a reference when designing its proposed allocation workflow.

Reference:https://www.notto.mohfw.gov.in/WriteReadData/Portal/News/848_1_NOTTO_TRANSPLANT_MANUAL-final_version.pdf

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4. Problem Statement and Objectives

Problem Statement: To develop a centralized and secure system for managing organ donation, compatibility matching, allocation, and communication between donors, recipients, and hospitals.

Problem Description: Manual coordination of organ transplantation can cause delays, difficulties in tracking organ availability, and communication problems. LifeBridge aims to organize these activities through a microservices-based platform.

Objectives:

1. To maintain donor, recipient, and organ details in a centralized system.
2. To support organ compatibility matching based on relevant medical information.
3. To organize the organ allocation process according to applicable guidelines.
4. To provide notifications about important transplant-related updates.
5. To improve data security through authentication and role-based access.
6. To develop a modular and scalable system using Java, Spring Boot, REST APIs, and MySQL.

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

- Processor: Intel Core i3 or above
- RAM: Minimum 8 GB
- Hard Disk: Minimum 10 GB of free space

- Internet Connection: Required for development and application access

Software Requirements:

- Operating System: Windows 10 or above
- Programming Language: Java
- Backend Framework: Spring Boot
- Database: MySQL
- API Technology: REST APIs
- Development Environment: Visual Studio Code
- Database Tool: MySQL Workbench
- Build Tool: Maven
- Web Browser: Google Chrome or Microsoft Edge

6. IMPLEMENTATION

1. Frontend: HTML, CSS and JavaScript

2. Backend: Java and Spring Boot

3. API Integration: REST APIs

1. Frontend (HTML, CSS, JavaScript):

- index.html: Main webpage for the application.
- login.html: User login page.
- register.html: Donor and recipient registration page.
- dashboard.html: Displays donor, recipient and organ information.
- style.css: Defines the design and layout of the web pages.
- script.js: Handles user interactions and frontend operations.

2. Backend (JAVA):

- LifeBridgeApplication.java: Main Spring Boot application file used to run the backend.
- DonorController.java: Handles donor-related operations.
- RecipientController.java: Handles recipient-related operations.

- OrganController.java: Manages organ information and availability.
- MatchingService.java: Handles organ compatibility matching logic.
- AllocationService.java: Manages the organ allocation process.
- NotificationService.java: Handles transplant-related notifications.
- Repository files: Manage communication between the application and MySQL database.
- Entity files: Define donor, recipient and organ data.

3. API Integration:

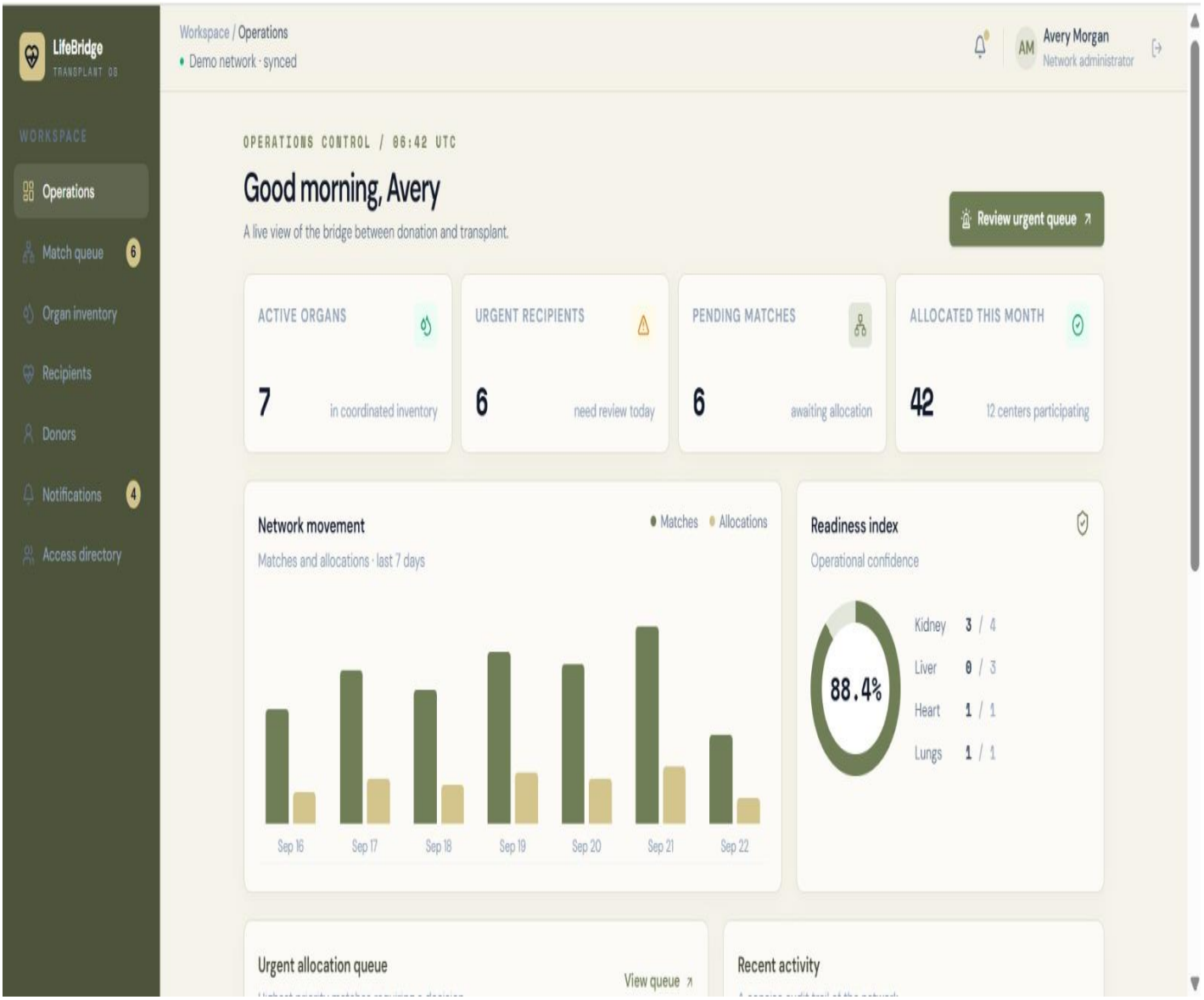
- Donor API: Handles donor registration and donor information.
- Recipient API: Manages recipient details.
- Organ API: Provides organ availability information.
- Matching API: Supports compatibility-based matching.
- Allocation API: Handles organ allocation requests.
- Notification API: Supports communication of important updates.

REST APIs are used to enable communication between the frontend and backend services. MySQL is used to store and retrieve application data.

7. RESULTS & DISCUSSION

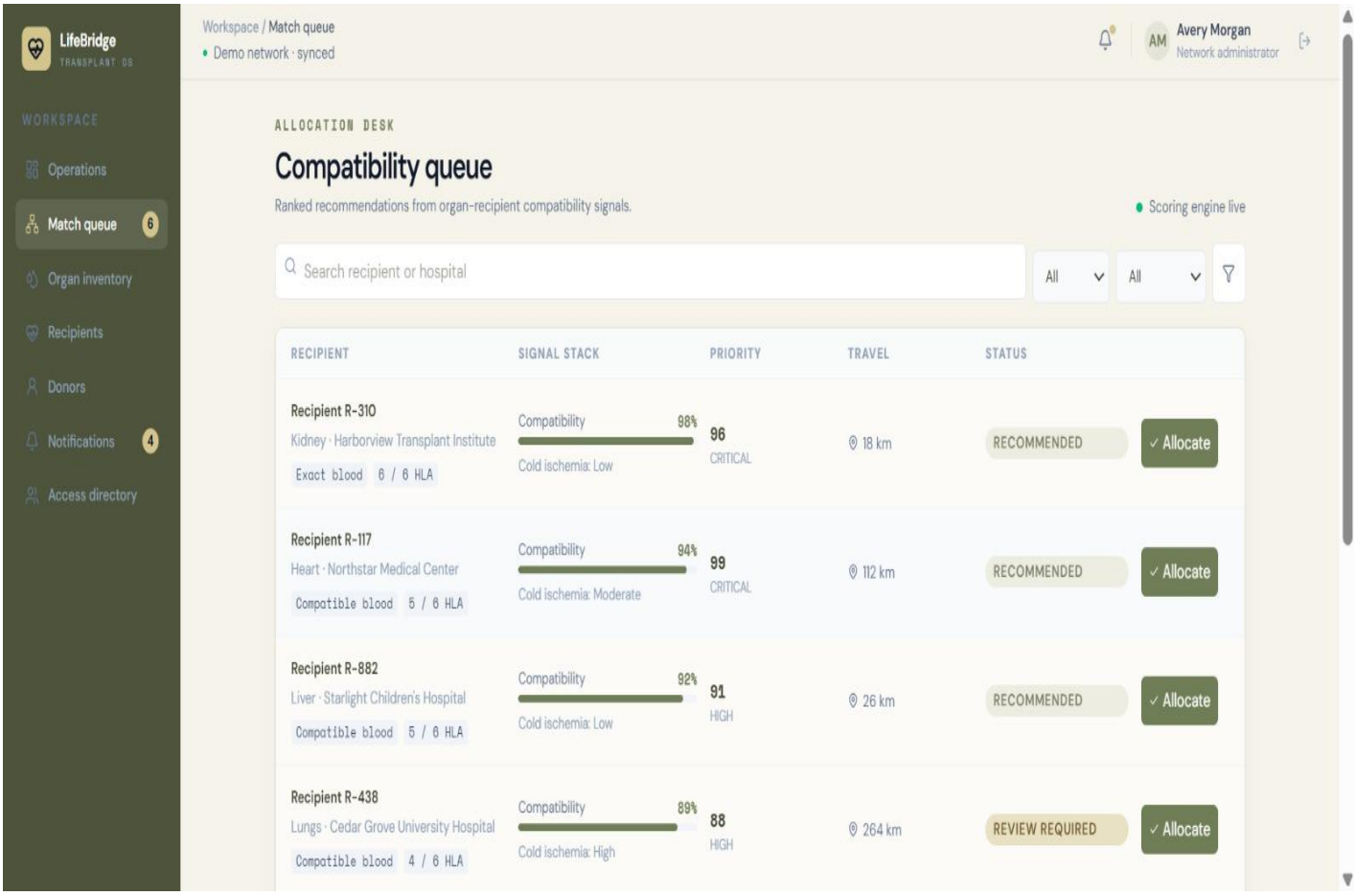
1. Operations Dashboard

The dashboard provides an overall view of the organ transplant management system, including active organs, urgent recipients, pending matches and completed allocations. It also displays network activity and organ readiness information for easy monitoring.


LifeBridge
TRANSPLANT OS
WORKSPACE
Operations
Match queue 6
Organ inventory
Recipients
Donors
Notifications 4
Access directory
Workspace / Operations
Demo network · synced
AM
Avery Morgan
Network administrator
OPERATIONS CONTROL / 06:42 UTC
Good morning, Avery
A live view of the bridge between donation and transplant.
Review urgent queue
ACTIVE ORGANS
7
in coordinated inventory
URGENT RECIPIENTS
6
need review today
PENDING MATCHES
6
awaiting allocation
ALLOCATED THIS MONTH
42
12 centers participating
Network movement
Matches
Allocations
Matches and allocations · last 7 days
Sep 16
Sep 17
Sep 18
Sep 19
Sep 20
Sep 21
Sep 22
Readiness index
Operational confidence
88.4%
Kidney 3 / 4
Liver 0 / 3
Heart 1 / 1
Lungs 1 / 1
Urgent allocation queue
View queue
Recent activity

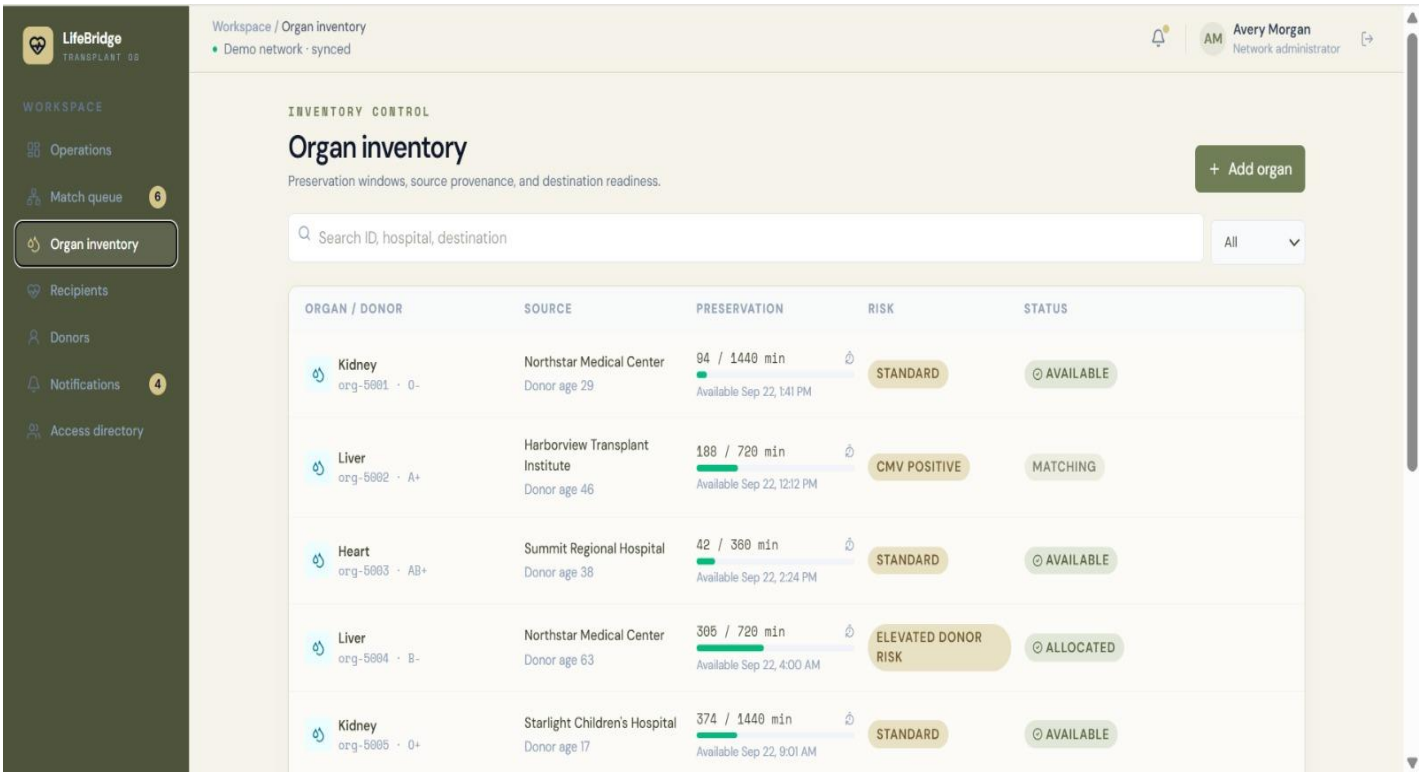
2. Compatibility Matching Queue

This page displays recipients ranked according to organ compatibility, priority and other matching information. It helps administrators review suitable matches and proceed with the allocation process.



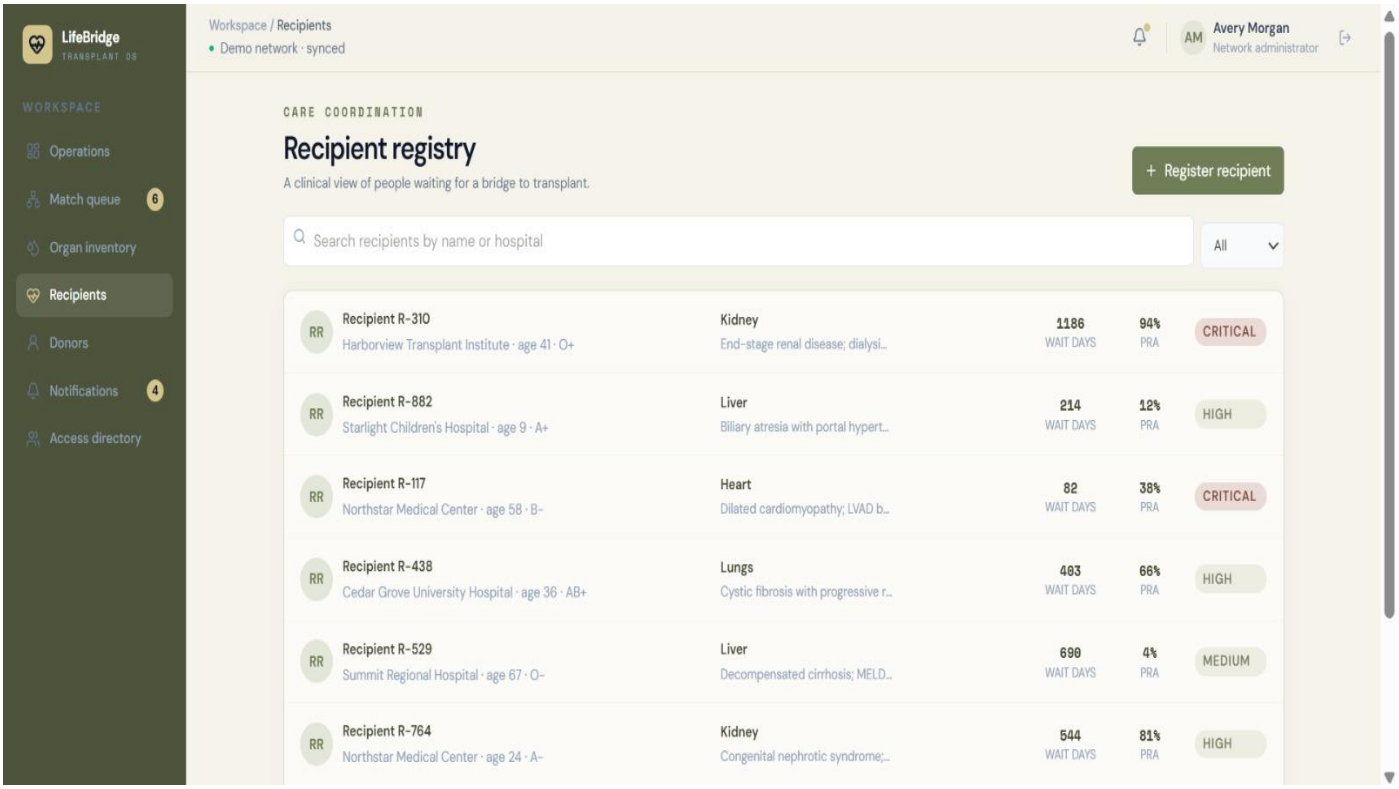
3. Organ Inventory

The organ inventory page displays available organs, donor details, source hospitals, preservation time, risk information and current status. It helps administrators track and manage organ availability



4. Recipient Registry

The recipient registry displays registered recipients along with their required organs, hospital details, waiting time and priority status. It helps administrators monitor recipient information and identify urgent cases.



8. **CONCLUSION & FUTURE SCOPE**

The LifeBridge Smart Organ Transplant System aims to simplify the organ donation and transplantation process through a centralized platform. It focuses on managing donor and recipient details, tracking organ availability, supporting compatibility-based matching, organ allocation and notifications. By using Java, Spring Boot, REST APIs and MySQL with a microservices architecture, the proposed system aims to improve coordination, security and efficiency in transplant management.

The future scope of LifeBridge includes developing a mobile application, introducing AI-based organ compatibility matching, providing real-time notifications and integrating with hospital management systems. Further enhancements may include blockchain technology for secure record management, advanced analytics and support for coordination between multiple hospitals and transplant centers.

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